

Section B

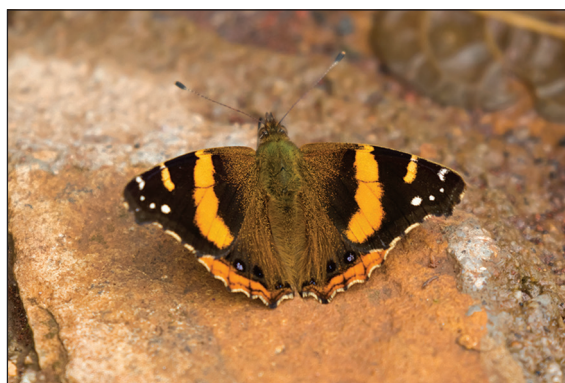
Refer to the separate Resource Folder to answer question 8.

8. (a) (i) The Red Admiral butterfly competes with greenfly for food. The scientific name for one type of greenfly is *Acyrtosiphon pisum*.

Complete the table below to show the scientific classification of greenfly. [3]

Kingdom:	
Phylum:	
Class:	Insecta
Order:	Hemiptera
Family:	Aphididae
Genus:	
Species:	
Scientific name:	<i>Acyrtosiphon pisum</i>

- (ii) Another butterfly is found in Africa. Its scientific name is *Vanessa abyssinica*.



- I. State **one** advantage of using scientific names. [1]

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- II. State **one** piece of evidence, from its scientific name, that shows it is related to the Red Admiral. [1]

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(b) The variation in population of adult Red Admirals is shown by the graph in **Figure 1**.

- (i) I. State the name of the seasonal movement of animals as seen in Red Admirals. [1]

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II. State **one** reason why this seasonal movement occurs. [1]

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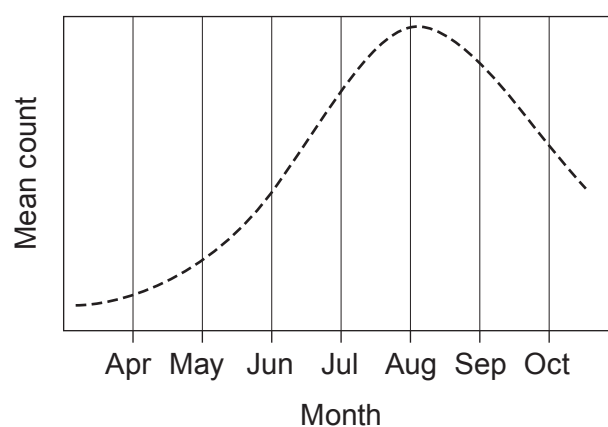
- (ii) Use the information in the graph in **Figure 1** to compare the variation of the Red Admiral population in 2019 with the mean change since 1976. [2]

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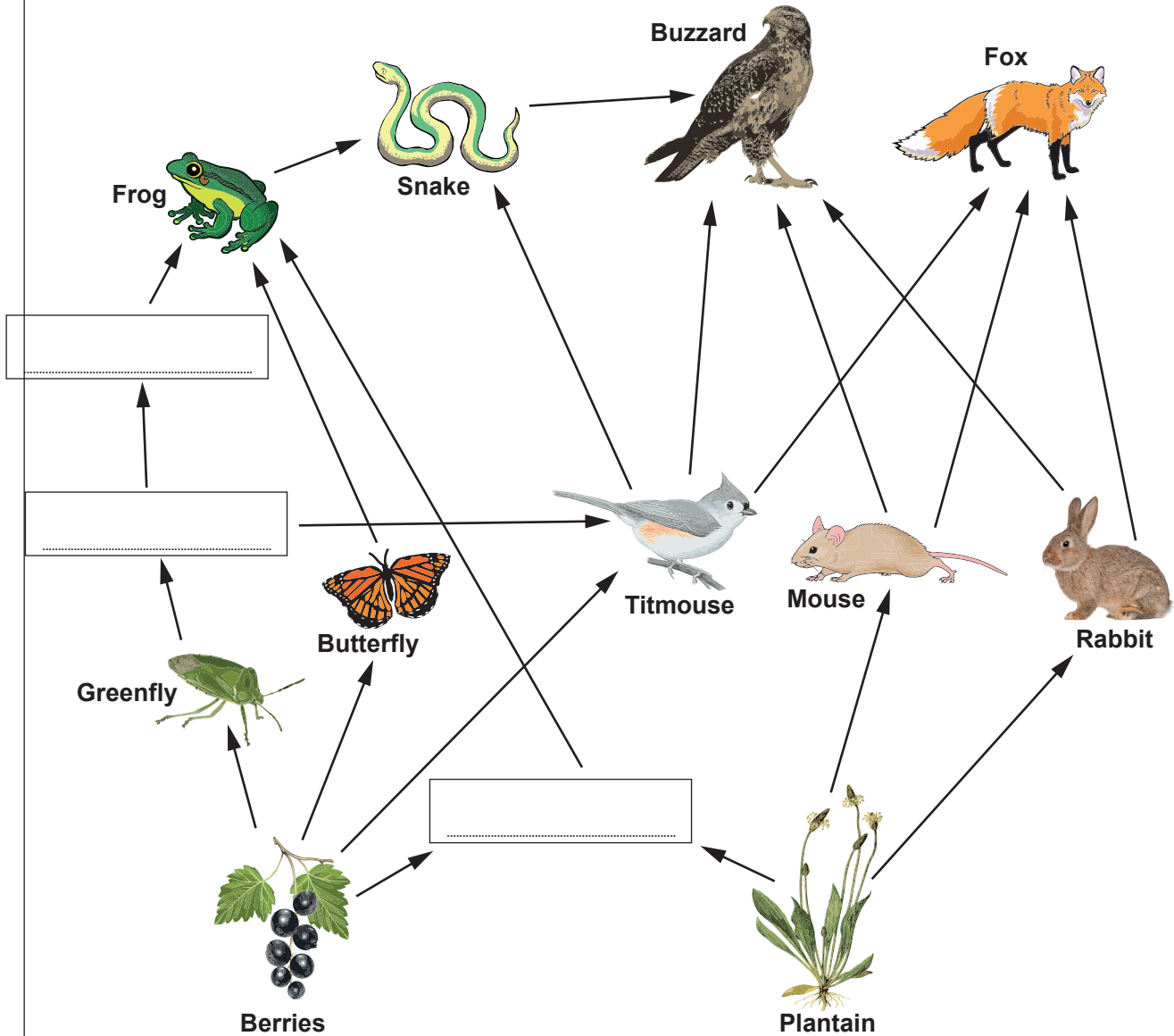
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- (iii) Red Admirals are a food source for frogs. **Add a line** to the graph below to show how the population of frogs changed between April and October in 2019. [2]



(c) **Table 2** gives information about the feeding relationships in one woodland habitat.

(i) Use the information in **Table 2** to **complete** the food web below. [3]



(ii) In this habitat, the fields containing plantain are cleared to make way for housing. Explain how this affects the foxes. [3]

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(iii) Explain why the frog can be found at more than one trophic level. [2]

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(d) Refer to **Table 2** to construct a pyramid of biomass for a food chain of **five** trophic levels starting with berries and ending with buzzards. [3]

(e) Use the information in **Figure 2** to calculate the efficiency of energy transfer between the primary and secondary consumers. [3]

% efficiency =

END OF PAPER



THE PERIODIC TABLE

Group

3 4 5 6 7 0

1 H Hydrogen 1 11 B Boron 5 12 C Carbon 6 14 N Nitrogen 7 16 O Oxygen 8 19 F Fluorine 9 20 Ne Neon 10 4 He Helium 2																	
7 Li Lithium 3 23 Na Sodium 11 39 K Potassium 19 86 Rb Rubidium 37 133 Cs Caesium 55 223 Fr Francium 87 9 Be Beryllium 4 24 Mg Magnesium 12 40 Ca Calcium 20 88 Sr Strontium 38 137 Ba Barium 56 226 Ra Radium 88 45 Sc Scandium 21 89 Y Yttrium 39 139 La Lanthanum 57 227 Ac Actinium 89 51 V Vanadium 23 93 Nb Niobium 41 181 Ta Tantalum 73 48 Ti Titanium 22 91 Zr Zirconium 40 179 Hf Hafnium 72 52 Cr Chromium 24 96 Mo Molybdenum 42 184 W Tungsten 74 59 Co Cobalt 27 103 Rh Rhodium 45 192 Ir Iridium 77 56 Fe Iron 26 101 Ru Ruthenium 44 190 Os Osmium 76 55 Mn Manganese 25 99 Tc Technetium 43 186 Re Rhenium 75 59 Ni Nickel 28 106 Pd Palladium 46 195 Pt Platinum 78 63.5 Cu Copper 29 108 Ag Silver 47 197 Au Gold 79 65 Zn Zinc 30 112 Cd Cadmium 48 201 Hg Mercury 80 70 Ga Gallium 31 115 In Indium 49 204 Tl Thallium 81 73 Ge Germanium 32 119 Sn Tin 50 207 Pb Lead 82 75 As Arsenic 33 122 Sb Antimony 51 209 Bi Bismuth 83 79 Se Selenium 34 128 Te Tellurium 52 210 Po Polonium 84 80 Br Bromine 35 127 I Iodine 53 210 At Astatine 85 84 Kr Krypton 36 131 Xe Xenon 54 222 Rn Radon 86																	

Key

relative atomic mass

Ar

Symbol

Name

Z

atomic number

